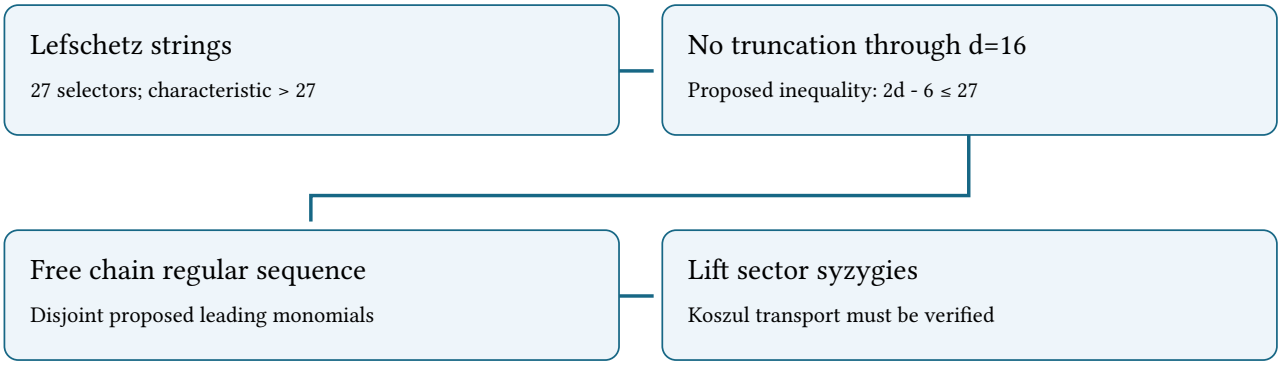
Candidate reduction of prime chain syzygies

Unvalidated research direction. No new FFD/LFD result or production run.

The cubic development maps expose the cost of direct coefficient matrices: the prospective m9 domain has882816columns. This search asks whether the original27-selector systems can be analyzed by graded modules instead. The existing conditional FFD intervals7–17 remain authoritative.



Literature checked. Lundqvist–Nicklasson,Theorem2.7 and Section3, gives the strong Lefschetz property for multiplication by the coordinate sum when the characteristic exceeds the socle degree. The block socle degree is27, below the83-bit prime. This establishes a block fact,not the chain argument.

Proposed chain argument. Decompose each squarefree block into graded strings. In a primitive sector of total degree K,syzygy coefficients have degree d-4-K. Chain quartics have degree at most two in each block sum. The proposed sufficient no-truncation condition is2d-6≤27,covering d≤16. Give block sums weight m and intermediate U_j weight m-2-j. The candidate leading monomials have pairwise disjoint supports. If these checks establish a regular sequence in the free block-sum ring,transport its Koszul syzygies through the strings and then through selector-square reduction.

Open obligations. Prove the graded decomposition and transport,check every boundary and source normalization,compare sector kernels against the retained direct maps,and include bad-characteristic/truncation/leading-term negative controls. Freeze any all42-case production certificate run separately. Existing upper17 evidence can only be joined after exact source/base identity checks. Independent scientific review and formalization remain pending.

No exactFFD17 claim is made for higher m here. No LFD,solving-degree or elite-base conclusion follows automatically. PRIME_LEFSCHETZ_CANDIDATE.md records the full proposed argument and evidence checklist. The direct cubic development data are preserved as reference controls.

Lundqvist–Nicklasson,On the structure of monomial complete intersections in positive characteristic. Caminata–Gorla,Solving degree,last fall degree,and related invariants.